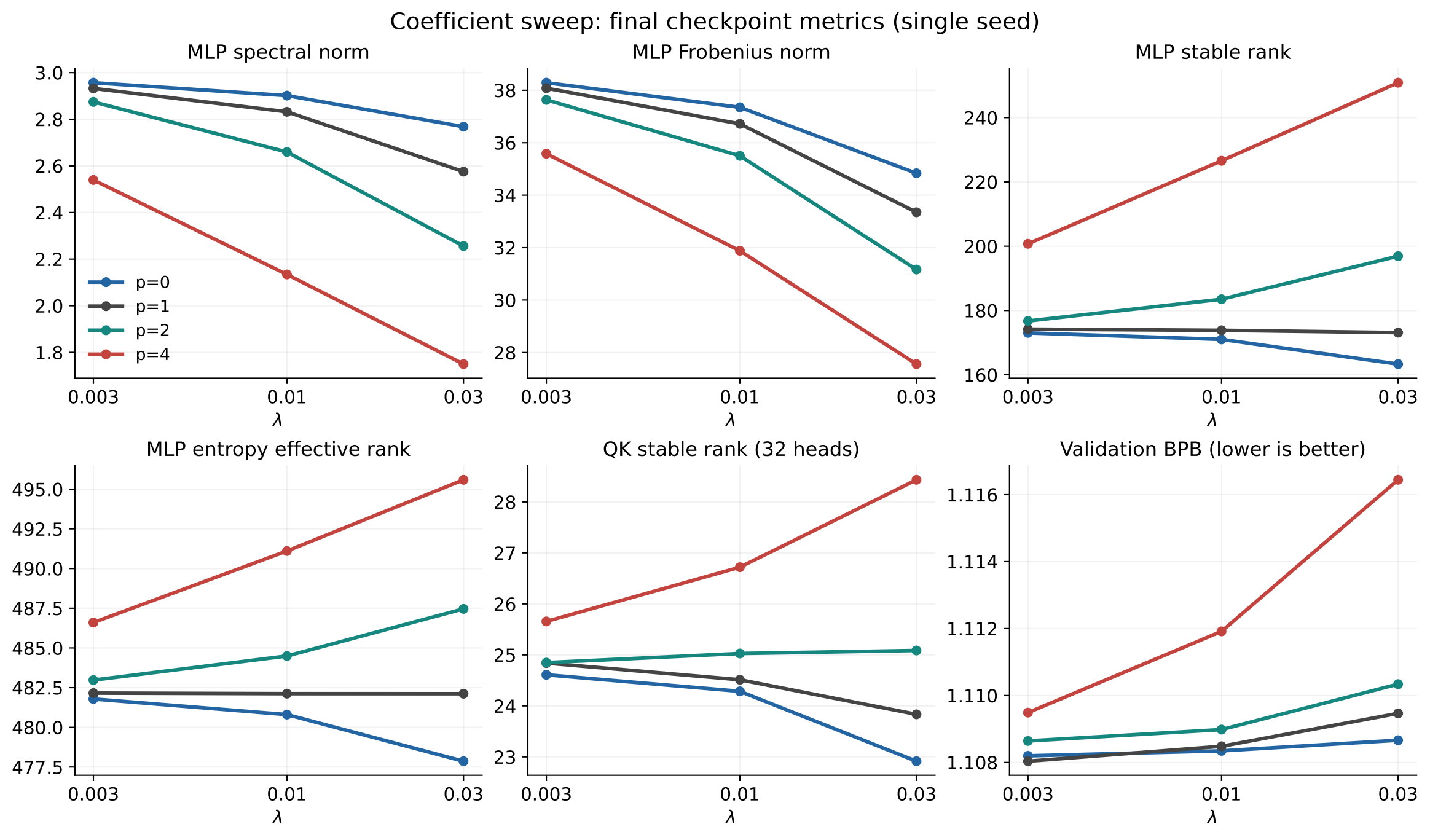
Coefficient sweep: final checkpoint metrics (single seed)
MLP spectral norm
MLP Frobenius norm
MLP stable rank
MLP entropy effective rank
QK stable rank (32 heads)
Validation BPB (lower is better)
p=0
p=1
p=2
p=4
λ
0.003
0.01
0.03